

Subject 3.

(Pb1)

$$\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \cdot f(x) dx = A_1 f(-1) + A_2 f(x_2) + A_3 f(x_3) + A_4 f(1) + R(f)$$

nodulele cotate fixate $\Rightarrow$ Formula Gauss Lobatto

$$x_0 = -1, \quad x_3 = 1$$

 $\Rightarrow x_1, x_2$ sunt zerourile lui $\pi_2(\cdot, w_{a,b})$

$$a = -1, \quad b = 1$$

$$\begin{aligned} w_{a,b}(x) &= (x-a)(b-x) \cdot w(x) \\ &= (x+1)(1-x) \cdot \frac{1}{\sqrt{1-x^2}} = \frac{1-x^2}{\sqrt{1-x^2}} \end{aligned}$$

$$\begin{aligned} w_{a,b}(x) &= \sqrt{1-x^2} \\ I &= [-1, 1] \end{aligned} \quad \rightarrow \text{polinomul ortogonal} \\ \text{cebiru 2.}$$

$$Q_m(x) = \frac{\sin((m+1) \cdot \arccos(x))}{\sqrt{1-x^2}}; \quad x \in [-1, 1]$$

$$x_k = \cos \frac{k \cdot \pi}{n}; \quad k = \overline{0, \dots, n}$$

$$\begin{cases} x_1 = \cos \frac{\pi}{3} = \frac{1}{2} \\ x_2 = \cos \frac{2\pi}{3} = -\frac{1}{2} \end{cases}$$

$$f \equiv 1: \quad A_1 + A_2 + A_3 + A_4 = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \pi$$

$$f \equiv x: \quad A_1 \cdot (-1) + A_2 \cdot \left(\frac{1}{2}\right) + A_3 \cdot \left(-\frac{1}{2}\right) + A_4 \cdot 1 = \int_{-1}^1 \frac{x}{\sqrt{1-x^2}} dx = 0$$

$$f \equiv x^2: \quad A_1 \cdot (-1)^2 + A_2 \cdot \left(\frac{1}{2}\right)^2 + A_3 \cdot \left(-\frac{1}{2}\right)^2 + A_4 = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2}$$

$$f \equiv x^3: \quad A_1 \cdot (-1)^3 + A_2 \cdot \left(\frac{1}{2}\right)^3 + A_3 \cdot \left(-\frac{1}{2}\right)^3 + A_4 = \int_{-1}^1 \frac{x^3}{\sqrt{1-x^2}} dx = 0$$

$$\Rightarrow \begin{cases} A_1 = 0,52 \\ A_2 = A_3 = 1,04 \\ A_4 = 0,52 = A_1 \end{cases}$$

$$R_m(f) = \frac{f^{(2m-2)}(\xi)}{(2m-2)!} \cdot \int_{-1}^1 w_2(x) \cdot Q_{m-2}^2(x) dx$$

$$= \frac{f^{(6)}(\xi)}{6!} \cdot \int_{-1}^1 \sqrt{1-x^2} \cdot \left(\frac{\sin(3 \arccos x)}{\sqrt{1-x^2}} \right)^2 dx$$

$$= \frac{f^{(6)}(\xi)}{6!} \cdot \int_{-1}^1 \sin(3 \arccos x) dx$$

$$= \frac{f^{(6)}(\xi)}{6!} \cdot \int_{-1}^1 \underbrace{\sqrt{1-x^2} \cdot \left(\frac{\sin(3 \arccos x)}{\sqrt{1-x^2}} \right)^2}_{\frac{u}{2}} dx$$

$$R_4(f) = \frac{u}{2} \cdot \frac{f^{(6)}(\xi)}{6!} ; \xi \in (-1, 1)$$

$$\begin{aligned} \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} f(x) dx &= 0,52 \cdot f(-1) + 1,04 \cdot f\left(\frac{1}{2}\right) + \\ &+ 1,04 \cdot f\left(-\frac{1}{2}\right) + 0,52 \cdot f(1) + \\ &\frac{u}{2} \cdot \frac{f^{(6)}(\xi)}{6!}, \xi \in (-1, 1) \end{aligned}$$

(Pb2)

a) Fie α un zero simplu al lui f și $f \in C^p$ în vecinătatea lui α , unde $p \geq 3$. Arătați că: dacă $f''(\alpha) = \dots = f^{(p-1)}(\alpha) = 0$, $f^{(p)}(\alpha) \neq 0$ atunci metoda lui Newton aplicată lui $f(x) = 0$ converge local către α cu ordinul p . Det. constanta de eroare asimptotică

M. lui Newton: $x_{m+1} = x_m - \frac{f(x_m)}{f'(x_m)}$

Rescriem pb. ca o pb de punct fix:

$$x_{m+1} = T(x_m)$$

Arătăm că: $T(x) = x \Leftrightarrow f(x) = 0$

$$x - \frac{f(x)}{f'(x)} = x \Leftrightarrow \frac{f(x)}{f'(x)} = 0 \Leftrightarrow f(x) = 0$$

Teorema: Fie α un punct fix al lui T și $I_\varepsilon = \{x \in \mathbb{R} : |x - \alpha| \leq \varepsilon\}$. Ip. că $f \in C^p[I_\varepsilon]$ și $\max_{t \in I_\varepsilon} |T'(t)| \leq 1$ at. (x_n) definit

$x_{m+1} = T(x_m)$ converge către α , $\forall x_0 \in I_\varepsilon$

$$\begin{aligned} T'(x) &= 1 - \frac{f'(x)^2 - f(x) \cdot f''(x)}{f'(x)^2} = 1 - 1 + \frac{f(x) \cdot f''(x)}{f'(x)^2} \\ &= \frac{f(x) \cdot f''(x)}{f'(x)^2} \end{aligned}$$

$$x = \alpha \in I_\varepsilon; |T'(\alpha)| = \left| \frac{f(\alpha) \cdot f''(\alpha)}{f'(\alpha)^2} \right| = 0 < 1 \quad (1)$$

1) $\Rightarrow$ convergența locală către α

$$\left[C = \frac{1}{p!} \cdot f^{(p)}(\alpha) \right]$$

C - constanta de eroare asimptotică

b) Deduceți o metodă pt. calculul lui $\sqrt[4]{a}$, aplicând a) funcției

$$f(x) = x^{4-m} - a \cdot x^{-m}$$

Cat este ordinul de convergență?

$$f(\sqrt[4]{a}) = a^{\frac{4-m}{4}} - a \cdot a^{-\frac{m}{4}} = a^{\frac{4-m}{4}} - a^{\frac{4-m}{4}} = 0$$

$$\Rightarrow \alpha = \sqrt[4]{a} \text{ răd. a ec. } f(x) = 0.$$

$$f'(x) = (4-m) \cdot x^{(3-m)} + a \cdot m \cdot x^{(-m-1)}$$

$$f''(x) = (4-m)(3-m) \cdot x^{(2-m)} + a \cdot m \cdot (-m-1) \cdot x^{(-m-2)}$$

Dim a) știm că: $f''(\alpha) = \dots = f^{(p-1)}(\alpha) = 0$
 $f^{(p)}(\alpha) \neq 0.$

$$\Rightarrow f''(\sqrt[4]{a}) = 0.$$

$$f''(\sqrt[4]{a}) = (4-m)(3-m) \cdot a^{\frac{2-m}{4}} + a \cdot m \cdot (-m-1) \cdot a^{\frac{-m-2}{4}}$$

$$\begin{aligned} f''(\sqrt[4]{a}) &= (4-m)(3-m) \cdot a^{\frac{2-m}{4}} + m(-m-1) \cdot a^{\frac{2-m}{4}} \\ &= a^{\frac{2-m}{4}} (12 - 4m - 3m + m^2 - m^2 - m) \\ &= a^{\frac{2-m}{4}} (12 - 8m) = 0 \end{aligned}$$

$$\text{Cum } a > 0 \Rightarrow 12 - 8m = 0$$

$$12 = 8m$$

$$m = \frac{12}{8} = \frac{3}{2}$$

$$m = \frac{3}{2}: f'(x) = \left(4 - \frac{3}{2}\right) \cdot \left(3 - \frac{3}{2}\right) \cdot x^{\left(2 - \frac{3}{2}\right)} + a \cdot \frac{3}{2} \cdot \left(-\frac{3}{2} - 1\right) \cdot x^{\left(-\frac{3}{2} - 1\right)}$$

$$\begin{aligned} f'(x) &= \frac{5}{2} \cdot \frac{3}{2} \cdot x^{\frac{1}{2}} + a \cdot \frac{3}{2} \cdot \left(-\frac{5}{2}\right) \cdot x^{-\frac{5}{2}} \\ &= \frac{15}{4} \cdot x^{\frac{1}{2}} - \frac{15}{4} \cdot a \cdot x^{-\frac{5}{2}} \end{aligned}$$

$$f''(\sqrt[4]{a}) = \frac{15}{4} \cdot a^{\frac{1}{8}} - \frac{15}{4} \cdot a^{1 - \frac{5}{8}} = \frac{15}{4} \cdot a^{\frac{1}{8}} - \frac{15}{4} \cdot a^{\frac{1}{8}} = 0$$

$$\begin{aligned} f''(x) &= \frac{15}{4} \cdot \frac{1}{2} \cdot x^{\left(\frac{1}{2} - 1\right)} - \frac{15}{4} \cdot a \cdot \left(-\frac{5}{2}\right) \cdot x^{\left(-\frac{5}{2} - 1\right)} \\ &= \frac{15}{4} \cdot \frac{1}{2} \cdot x^{-\frac{1}{2}} + \frac{15}{4} \cdot a \cdot \frac{5}{2} \cdot x^{-\frac{9}{2}} \end{aligned}$$

$$\begin{aligned} f''(\sqrt[4]{a}) &= \frac{15}{8} \cdot a^{-\frac{1}{8}} + \frac{15 \cdot 5}{8} \cdot a^{1 - \frac{9}{8}} \\ &= \frac{15}{8} \cdot a^{-\frac{1}{8}} + \frac{15 \cdot 5}{8} \cdot a^{-\frac{1}{8}} = \frac{15 \cdot 8}{8} \cdot a^{-\frac{1}{8}} \\ &= 15 \cdot a^{-\frac{1}{8}} \neq 0 \Rightarrow \boxed{\text{ordinul de conv} = 3} \end{aligned}$$

Dim a) +) știm că N. Newton aplicată lui $f(x) = 0$, converge la $\alpha = \sqrt[4]{a}$ cu ordinul $p = 3$.

$$\bullet X_{m+1} = X_m - \frac{f(X_m)}{f'(X_m)}$$

- Criteriul de oprire: $|X_{m+1} - X_m| < \varepsilon$, ε -dat
- Alegem x_0 a.c. $f(x_0) \cdot f''(x_0) > 0$.

$$m = \frac{3}{2}: f'(x) = \left(4 - \frac{3}{2}\right) \cdot x^{\left(3 - \frac{3}{2}\right)} + a \cdot \frac{3}{2} \cdot x^{\left(-\frac{3}{2} - 1\right)}$$

$$f'(x) = \frac{5}{2} \cdot x^{\frac{3}{2}} + a \cdot \frac{3}{2} \cdot x^{-\frac{5}{2}} > 0, \quad \forall x > 0$$

pt. c. $f' > 0 \Rightarrow f$ este crescătoare;
 $f(\sqrt[4]{a}) = 0 \Rightarrow$ pt. c. $f(x_0) > 0, x_0 > \sqrt[4]{a}$

$$\sqrt[4]{a} = \sqrt[4]{a \cdot 1 \cdot 1 \cdot 1} \leq \frac{a + 1 + 1 + 1}{4} = \frac{a+3}{4}$$

$$\Rightarrow x_0 = \frac{a+3}{4}$$

$$f''(x_0) = \frac{15}{4} \cdot x_0^{\frac{1}{2}} - \frac{15}{4} \cdot a \cdot x_0^{-\frac{3}{2}} > 0, \forall a > 0$$

(grafic matlab)

```
%Exemplu apel pentru Newton
```

```
m=3/2;
```

```
x=sym('x');
```

```
a=16
```

```
f= x^(4-m)-a*x^(-m);
```

```
fd=diff(f);
```

```
x0=(a+3)/4;
```

```
precizie=10^(-15);
```

```
Nmax=100;
```

```
[z,ni]=NewtonEc(matlabFunction(f),matlabFunction(fd),x0,precizie,Nmax)
```

```

function [z,ni]=NewtonEc(f,fd,x0,precizie,Nmax)
%NEWTONS - metoda lui Newton pentru ecuatii in R
%Intrare
%f - functia
%fd - derivata
%x0 - valoarea de pornire
%precizie - precizia ceruta
%Nmax - numar maxim de iteratii
%Iesire
%z - aproximatia solutiei
%ni - numar de iteratii

xv=x0;
for k=1:Nmax
    xc=xv-f(xv)/fd(xv);
    if abs(xc-xv)<precizie %succes
        z=xc;
        ni=k;
        return
    end
    xv=xc; %pregatesc iteratia urmatoare
end
%esec
error('S-a depasit numarul maxim de iteratii')

```